TEXORA · LECTURE SHEET

Correlation and Regression

Scatter diagrams, Karl Pearson coefficient of correlation and its properties, the two regression lines and regression coefficients, prediction from a fitted line, and Spearman rank correlation.

STAT-2109 · PROBABILITY AND STATISTICS · TOPIC 8 · LEVEL 2 TERM 1

1. Bivariate data and the scatter diagram

Earlier topics summarised one variable at a time. Many real questions involve two variables measured together on the same units: advertising spend and sales, fibre length and yarn strength, machine speed and fault rate. Such paired observations (x, y) are called **bivariate data**, and the first step in studying them is always a picture.

A **scatter diagram** plots each pair as a point, with the explanatory variable x on the horizontal axis and the response variable y on the vertical axis. The pattern of points reveals at a glance whether the variables move together (positive association), move oppositely (negative association), or show no linear pattern at all, and it also exposes curved patterns and isolated outliers that a single number would hide. Three archetypes are shown in Figure 1.

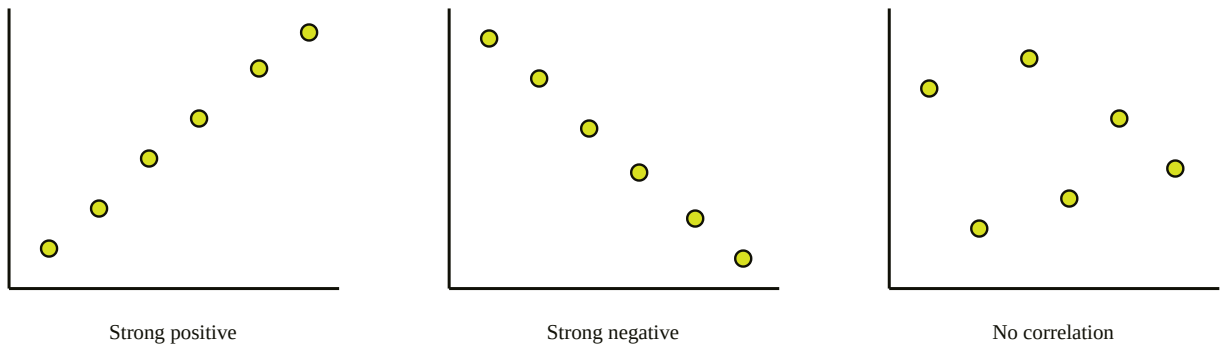


Figure 1 — Scatter diagrams: strong positive correlation, strong negative correlation, and no correlation

Two cautions belong with the picture. First, correlation in this topic means **linear** association: points arranged on a perfect curve can still give a correlation near zero, because no straight line summarises them. Second, association is not causation. Two variables may move together because a third variable drives both, or by coincidence in a small sample. The statistics of this sheet quantify association and support prediction; causal claims need design and subject knowledge beyond the numbers.

2. Karl Pearson coefficient of correlation

Karl Pearson product moment coefficient of correlation, denoted r, compresses the scatter into a single number. Writing $dx = x - \bar{x}$ and $dy = y - \bar{y}$, its defining form is

$$r = \Sigma(x - \bar{x})(y - \bar{y}) \,/\, \sqrt{[\ \Sigma(x - \bar{x})^2 \cdot \Sigma(y - \bar{y})^2\ ]}$$

The numerator is the sum of cross products of deviations. When large x values pair with large y values, the products dx·dy are mostly positive and r is positive; when large x pairs with small y, the products are mostly negative and r is negative. The denominator standardises the numerator to its maximum possible magnitude, which produces the famous bounds.

Properties of r.

- **$-1 \le r \le 1$** always. r = +1 means perfect positive linear relation, r = −1 perfect negative linear relation, and r = 0 no linear relation.
- **r is unit free.** It is a pure number, unchanged if x or y is converted to different units, because the units cancel between numerator and denominator.
- **r is symmetric:** the correlation of x with y equals the correlation of y with x. Correlation does not designate a cause and an effect.
- **r is unchanged by shifts of origin** and by positive changes of scale applied to either variable.
- **r measures linear association only**, so it must always be read alongside the scatter diagram.

As a verbal guide, |r| above about 0.9 is very strong, 0.7 to 0.9 is strong, 0.4 to 0.7 is moderate, and below 0.4 is weak, though the context of the subject always matters more than any fixed verbal scale.

3. Worked example: advertising and sales

A firm records advertising spend x and sales y, both in thousands of taka (000 Tk), for five periods. The computation is laid out in full because the same table also feeds the regression work of Section 4.

X	Y	DX = X − 30	DY = Y − 53	DX·DY	DX^2	DY^2
10	25	−20	−28	560	400	784
20	45	−10	−8	80	100	64
30	50	0	−3	0	0	9
40	65	10	12	120	100	144
50	80	20	27	540	400	729
Σx = 150	**Σy = 265**			**1300**	**1000**	**1730**

The means are $\bar{x} = 150 / 5 = 30$ and $\bar{y} = 265 / 5 = 53$. The column totals are $\Sigma(x - \bar{x})(y - \bar{y}) = 1300$, $\Sigma(x - \bar{x})^2 = 1000$ and $\Sigma(y - \bar{y})^2 = 1730$. Therefore

$$r = 1300 / \sqrt{(1000 \times 1730)} = 1300 / 1315.3 = 0.988$$

WORKED EXAMPLE

Advertising x (000 Tk): 10, 20, 30, 40, 50; sales y (000 Tk): 25, 45, 50, 65, 80.

$\bar{x} = 30$, $\bar{y} = 53$; $\Sigma dx \cdot dy = 1300$; $\Sigma dx^2 = 1000$; $\Sigma dy^2 = 1730$.

$r = 1300 / \sqrt{(1000 \times 1730)} = 1300 / 1315.3 =$ **0.988** — a very strong positive linear correlation between advertising spend and sales.

The value $r = 0.988$ is close to the maximum of 1: in these five periods, sales rose almost in lockstep with advertising. The square $r^2 \approx 0.977$ is often reported alongside r; it is interpreted in Section 5. Remember the caution of Section 1: the number shows association in the observed range, not proof that advertising alone caused the sales, and it licenses prediction only near the observed range of x.

4. Regression: the two regression lines

Correlation treats the two variables symmetrically. **Regression** breaks the symmetry: it uses one variable to predict the other through a fitted straight line, the line that minimises the sum of squared vertical prediction errors (the least squares principle). Because prediction can run in either direction, there are **two regression lines**:

$$\text{Regression of y on x:} \quad y - \bar{y} = b_{yx}\,(x - \bar{x}), \quad b_{yx} = \Sigma dx \cdot dy / \Sigma dx^2$$

$$\text{Regression of x on y:} \quad x - \bar{x} = b_{xy}\,(y - \bar{y}), \quad b_{xy} = \Sigma dx \cdot dy / \Sigma dy^2$$

The quantities b_{yx} and b_{xy} are the **regression coefficients**. The coefficient b_{yx} is the slope of the y on x line: the predicted change in y for a one unit increase in x, expressed in y units per x unit. Unlike r, a regression coefficient carries units and is not symmetric; swapping the roles of the variables gives a different line, except in the perfect correlation case.

WORKED EXAMPLES

From the advertising and sales table: $b_{yx} = 1300 / 1000 =$ **1.30** and $b_{xy} = 1300 / 1730 =$ **0.751**.

Line of y on x: $y - 53 = 1.30\,(x - 30)$, that is $\mathbf{y = 1.30x + 14}$.

Line of x on y: $x - 30 = 0.751\,(y - 53)$, that is $\mathbf{x = 0.751y - 9.83}$.

Prediction: at $x = 60$, the y on x line gives $\hat{y} = 1.30 \times 60 + 14 =$ **92** (000 Tk).

The slope 1.30 says that, within this data pattern, each additional thousand taka of advertising is associated with about 1.30 thousand taka of additional sales. The prediction at x = 60 thousand taka is sales of 92 thousand taka. Note that x = 60 lies just beyond the observed range of 10 to 50, so the prediction is a mild extrapolation and should be reported with that reservation; far extrapolation assumes the linear pattern persists where no data have been seen, which is often false.

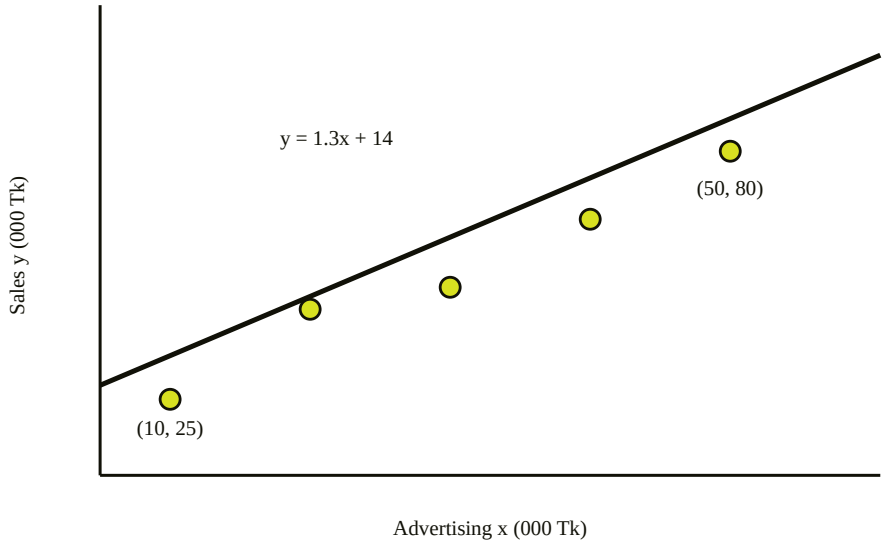


Figure 2 — Scatter of the advertising and sales data with the fitted regression line y = 1.3x + 14; the line passes through (10, 27) and (50, 79) and through the mean point (30, 53)

5. Properties connecting correlation and regression

- **Both regression lines pass through the mean point ($\bar{x}$, $\bar{y}$)**, here (30, 53). Their intersection is always the centre of the data, which is a quick check on any drawn or computed line.
- **The product of the regression coefficients equals r^2:** $b_{yx} \times b_{xy} = r^2$. Check with the worked example: $1.30 \times 0.751 = 0.977$, and $r^2 = 0.988^2 \approx 0.977$. The identity holds, confirming the arithmetic of both calculations at once.
- **r is the geometric mean of the two regression coefficients**, with the common sign of the coefficients: $r = \pm\sqrt{(b_{yx}\, b_{xy})}$.
- **If r = ±1, the two lines coincide** into a single line, because every point lies exactly on it and prediction is perfect in both directions.
- **If r = 0, the two lines are perpendicular**: the y on x line is horizontal (best prediction of y is simply $\bar{y}$, whatever x is) and the x on y line is vertical.
- Between the extremes, the closer r is to ±1, the closer the two lines lie to each other; the angle between them is a visual measure of the strength of correlation.

The quantity r^2, here 0.977, is called the **coefficient of determination**. It is read as the proportion of the variation in y accounted for by the linear relation with x: about 97.7% of the variation in sales across these five periods is accounted for by variation in advertising spend, leaving about 2.3% to all other factors and chance. This is a statement about variation in the sample, not a proof of causation.

6. Spearman rank correlation

Sometimes the data are ranks rather than measurements: judges orderings of fabric handle, grades of finish quality, or positions in a merit list. **Spearman rank correlation coefficient** applies Pearson formula to the ranks. When there are no tied ranks, it simplifies to

$$R = 1 - 6\,\Sigma d^2 / [\, n\,(n^2 - 1)\,]$$

where d is the difference between the two ranks given to the same item and n is the number of items. Like r, Spearman R lies between −1 and +1: R = +1 means the two rankings agree completely, R = −1 means one ranking is the exact reverse of the other, and R near 0 means no consistent agreement. Rank correlation is also useful for measured data when the relation is monotonic but not straight, or when outliers would distort Pearson r, because ranks limit the influence of any single extreme value. Tied ranks are handled by assigning average ranks and using the Pearson formula on the ranks directly; the simplified formula above assumes no ties.

Choosing between the coefficients is a matter of data type: Pearson r for measured pairs with a roughly linear scatter, Spearman R for ranks, ordinal judgements, or monotonic patterns. Both are reported with the sample size, and both deserve the same caution that association alone never proves causation.

Glossary

TERM	MEANING
Bivariate data	Paired observations (x, y) of two variables measured on the same units.
Scatter diagram	A plot of the paired observations as points, used to judge the form and strength of association.
Correlation	The degree of linear association between two variables, summarised by Pearson r.
Pearson r	$r = \Sigma dx \cdot dy / \sqrt{(\Sigma dx^2\ \Sigma dy^2)}$; a unit free number between −1 and +1.
Regression line	The least squares straight line used to predict one variable from the other.

Regression coefficient	The slope of a regression line: $b_{yx} = \Sigma dx \cdot dy / \Sigma dx^2$ for y on x, and $b_{xy} = \Sigma dx \cdot dy / \Sigma dy^2$ for x on y.
Coefficient of determination	r^2, the proportion of variation in y accounted for by the linear relation with x.
Spearman R	Correlation computed on ranks: $R = 1 - 6\Sigma d^2 / [n(n^2 - 1)]$ when there are no ties.

Key points & summary

- Always draw the scatter diagram first: it shows direction, strength, curvature and outliers that a single coefficient cannot.
- Pearson r satisfies $-1 \le r \le 1$, is unit free and symmetric, and measures linear association only.
- Worked example: $\Sigma dx \cdot dy = 1300$, $\Sigma dx^2 = 1000$, $\Sigma dy^2 = 1730$, giving $r = 1300 / 1315.3 = 0.988$, a very strong positive correlation.
- Regression coefficients are $b_{yx} = 1.30$ and $b_{xy} = 0.751$, giving the lines $y = 1.30x + 14$ and $x = 0.751y - 9.83$.
- Both lines intersect at the mean point (30, 53); they coincide when $r = \pm 1$ and are perpendicular when $r = 0$; $b_{yx} \times b_{xy} = r^2 = 0.977$.
- Prediction uses the correct directional line: at $x = 60$, $\hat{y} = 92$ (000 Tk), with an extrapolation caution since 60 lies beyond the observed range.
- Spearman R applies correlation to ranks and is used for ordinal data or monotonic relations.

Practice

Answer in your own words; brief model answers follow.

1. Sketch in words the three scatter patterns of strong positive, strong negative and zero correlation, and state what each implies about the variables.
2. State the formula for Pearson r and list three of its properties.
3. Using $\bar{x} = 30$, $\bar{y} = 53$, $\Sigma dx \cdot dy = 1300$, $\Sigma dx^2 = 1000$ and $\Sigma dy^2 = 1730$, compute r for the advertising and sales data and interpret the value.
4. Find both regression coefficients for the same data and write down both regression lines.
5. Use the appropriate line to predict sales when advertising spend is 60 (000 Tk), and state one caution about this prediction.

6. Verify the identity $b_{yx} \times b_{xy} = r^2$ for this data, state where the two lines intersect, and say what happens to the lines when r = 0 and when r = 1.

Answer key

1. Strong positive: points rise steeply from lower left to upper right in a narrow band, so large x goes with large y. Strong negative: points fall steeply from upper left to lower right, so large x goes with small y. Zero correlation: points form a shapeless cloud with no tilt, so knowing x gives no linear information about y.
2. $r = \Sigma(x - \bar{x})(y - \bar{y}) / \sqrt{[\Sigma(x - \bar{x})^2 \, \Sigma(y - \bar{y})^2]}$. Properties (any three): $-1 \le r \le 1$; unit free; symmetric in x and y; unchanged by change of origin; measures linear association only.
3. $r = 1300 / \sqrt{(1000 \times 1730)} = 1300 / 1315.3 = 0.988$. This is a very strong positive linear correlation: sales increased almost in step with advertising spend over the five periods.
4. $b_{yx} = 1300 / 1000 = 1.30$; $b_{xy} = 1300 / 1730 = 0.751$. Line of y on x: $y - 53 = 1.30 (x - 30)$, that is $y = 1.30x + 14$. Line of x on y: $x - 30 = 0.751 (y - 53)$, that is $x = 0.751y - 9.83$.
5. Use the y on x line: $\hat{y} = 1.30 \times 60 + 14 = 92$, so predicted sales are 92 (000 Tk). Caution: x = 60 lies outside the observed range 10 to 50, so this is an extrapolation that assumes the linear pattern continues beyond the data.
6. $1.30 \times 0.751 = 0.977$, and $r^2 = 0.988^2 \approx 0.977$, so the identity holds. The lines intersect at the mean point (30, 53). When r = 0 the lines are perpendicular (horizontal and vertical through the mean point); when r = 1 they coincide in a single line.

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